

A Hybrid Method for Critical Clearing Time Evaluation of Multi-machine Systems

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Abstract—This paper proposes a hybrid Critical Clearing Time(CCT) evaluation method. This method is based on an integrated approach of Bisection and Single Machine Equivalent (SIME) technique. In this method initially a time interval in which actual CCT lies is identified followed by applying bisection method for estimation of exact CCT. The method uses SIME based rotor-speed early stop criteria for detection of first-swing stability, thus reducing the computation time required. The hybrid technique is applied to WSCC 9 bus, IEEE 14 bus, IEEE 30 bus and New England 39 bus test system. The obtained CCT results are compared with the results of conventional Time Domain Simulation (TDS) approach. The proposed Bisection and SIME based integrated approach computes the CCT faster compared to conventional TDS approach.

Keywords—Critical Clearing Time(CCT); One Machine Infinite Bus(OMIB); Single Machine Equivalent(SIME); Time Domain Simulation(TDS); Transient Stability Assessment(TSA).

I. INTRODUCTION

The power system transient stability assessment is very important in power system planning and system studies. Transient stability problems caused by large disturbances are of concern for secure operation of power system. Transient stability is the ability of the power system to maintain synchronism after subjected to a large disturbance such as a short circuit on a transmission line [1]. Modern power systems being complex are operated under pressure. The operating stability limit evaluation is an important issue for present day competitive deregulated environment. The Critical Clearing Time (CCT) gives the boundary of transient stability for a particular fault condition. When the power system is subjected to a large disturbance, CCT distinguishes the operating condition as stable or unstable. It is very essential to estimate the CCT in order to avoid cascading outages or blackouts due to occurrence of large disturbances which may result in large excursions of rotor angles of synchronous machines leading to loss of synchronism.

The transient stability assessment methods are generally classified into time domain simulation, direct and automatic learning approaches. In time domain simulation method the nonlinear equations are solved simultaneously. The TDS approach simulates the system dynamics in the during-fault and post-fault configurations. The during-fault time period is very small whereas post-fault period may be much longer leading to a very time consuming CCT evaluation process. The estimation of CCT using Liapunov's type energy function methods which

are also called as direct methods or transient energy function methods have been discussed in [1]-[4]. In these methods CCT or stability of the system is assessed by calculation of pre and post-fault energy stored in the system. But for these methods there is a need for computation of accurate critical energy which is a difficult task and requires greater computation. Automatic learning (AL) approaches used for TSA are of two types namely, supervised and unsupervised which requires long-time for off-line training process. In [5] a transient energy based direct method for evaluation of transient stability is discussed. In literature [6] CCT is computed using an integrated approach of Potential Energy Boundary Surface method and time domain simulation method. Initially an approximate CCT is estimated using energy function approach and is used as a starting fault clearing time for time domain simulation to evaluate accurate CCT. In [7] the authors propose an energy comparison method for finding the severely disturbed machines (SDM) and hence to find CCT based on extension of equal area criterion for transient stability analysis of multi-machine power systems. In [8] a radial basis function network (RBFN) based ANN method is used for estimation of CCT. An online transient stability prediction scheme based on live synchrophasor data is proposed in [9]. Here the transient stability is predicted based on only few sample fault data received by PMUs without solving computationally extensive electromechanical dynamics. In [10] a critical trajectory based minimization problem is formulated for evaluation of CCT. The authors in [11] propose two new methods based on clustering identification for real time computation of CCT. The CCT is estimated using power angle and acceleration curves by transformation of multi-machine system into single machine equivalent. An algorithm for deriving a uniform transient stability index using single time, time-domain simulation is proposed in [12]. In [13] a method for transient stability assessment of the stable cases using the deceleration area of the power-angle representation of SIME method is proposed. A SIME method based algorithm for integration with standard time domain simulation program to search for stability limits is proposed in [14]. In literature [15] an educational graphical user interface tool for evaluation of multi-machine transient stability combining the SIME method with MATLAB and Simulink is described.

In this paper a SIME based hybrid method for transient stability analysis is presented. Consider the phase plane trajectories of an OMIB system as shown in figure 1. Three different types of trajectories corresponding to a 'stable' case, a 'critically stable' case and an 'unstable' case have been shown.

In a single machine equivalent system, for a stable case where the fault is cleared before the CCT the trajectory spirals inwards, towards the stable equilibrium point (SEP). For a critically stable case where the fault is cleared just before the CCT, the fault-on trajectory reaches the unstable equilibrium point (UEP) and starts spiraling around the SEP. For an unstable case where the fault is cleared after the CCT, the fault-on trajectory before reaching UEP goes away from SEP. UEP is the point where the generators in the system start losing the synchronism. From the figure 1 it can be seen that for a stable case the omega or speed deviation of OMIB passes by zero-axis and for an unstable case omega will not pass through zero-axis after the fault is cleared. Using this criteria one can classify the contingencies as stable or unstable for a particular fault clearing time. In this paper this criteria is used as early stability detection criteria for stopping integration process. Bisection method is used as an iterative technique for searching accurate CCT using the above mentioned stability detection criteria.

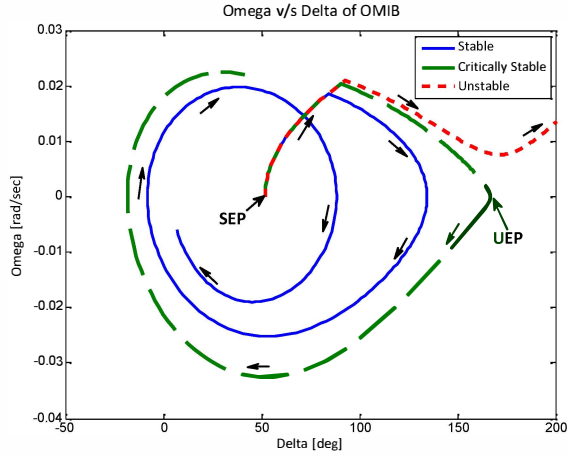


Fig. 1. Phase plane trajectories of the OMIB system

II. PROPOSED METHOD

The conventional time domain simulation method used for stability assessment requires more computation time for assessing the stability. In regular TDS based CCT evaluation process fault clearing time is a very important parameter which is an unknown quantity for a particular contingency. In conventional CCT evaluation process, for a given contingency the fault clearing time starts from a very low value and then it is gradually increased until the system becomes unstable. When for a particular fault clearing time the system becomes unstable that gives the CCT for that contingency. For each fault clearing time, in order to assess the stability of the system the integration needs to be carried out up to a few seconds. This process requires a very long computation time. In order to reduce this computation time required for repetitive analysis, bisection method is used for finding the accurate fault clearing time. A SIME based OMIB speed deviation criterion is used for stability detection. For a stable case as shown in figure 2, the speed deviation of the OMIB starts decreasing and crosses the zero-axis in few time steps after the fault is cleared [4]. But for an unstable case the speed starts increasing indicating

acceleration of the machines and will not cross zero-axis as shown in figure 3. This SIME based criteria for stability detection requires very less time as compared to conventional TDS method. The proposed method has been tested on WSCC 9 bus system, IEEE 14 bus, IEEE 30 bus and 39 bus New England test systems. The CCT evaluated using proposed method matches with the results obtained by conventional TDS method.

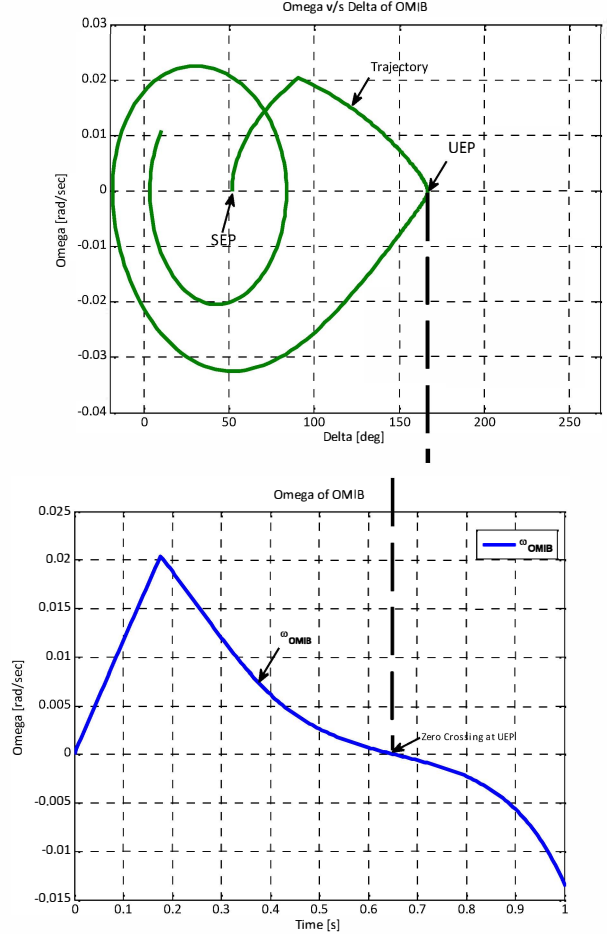


Fig. 2. Fault-on trajectory and the corresponding speed of OMIB for a stable case

III. MATHEMATICAL MODEL

A. Power System Model

In a multi-machine power system, the dynamic behavior of an i^{th} generator can be described by the following equations:

$$\frac{H_i}{\pi f} \frac{d\omega_i}{dt} = P_{mi} - P_{ei} \dots (1)$$

$$\frac{d\delta_i}{dt} = \omega_i \dots (2)$$

where:

P_{ei} – Electrical power output of machine i in pu.

$$P_{ei} = \sum_{j=1}^n |E'_i| |E'_j| |Y_{ij}| \cos(\theta_{ij} - \delta_i + \delta_j)$$

P_{mi} – Mechanical power output of machine i in pu.

ω_i – Speed deviation of machine i in rad per sec.

δ_i – Rotor angle of machine i in radians.

H – Inertia constant in MW-s/MVA.

E' – Generator voltage behind the transient reactance in pu.

Y_{ij} – Admittance between the nodes i and j in pu.

f – System frequency in Hz.

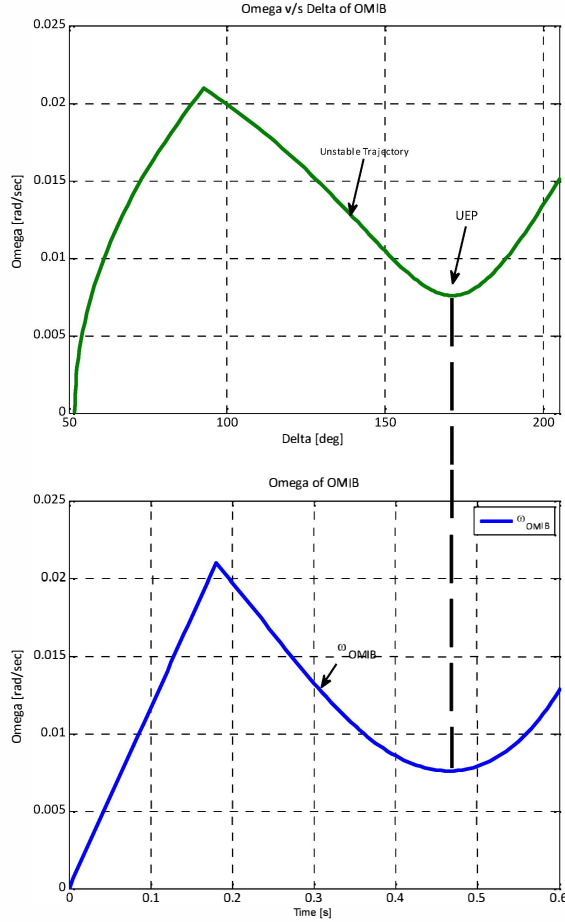


Fig. 3. Fault-on trajectory and corresponding speed of OMIB for an unstable case

The rotor angles and speed thus obtained can be converted from synchronous reference frame to center-of-inertia (COI) or center-of-angle (COA) reference frame using the following equations.

$$\left. \begin{aligned} \theta_i &= \delta_i - \delta_0 \\ \tilde{\omega}_i &= \dot{\theta}_i = \omega_i - \omega_0 \end{aligned} \right\} \dots\dots (3)$$

where:

θ_i – Angular displacement relative to COA

$\tilde{\omega}_i$ – Angular velocity relative to COA

$$\delta_0 = \frac{1}{M_T} \sum_{i=1}^n M_i \delta_i \quad \omega_0 = \frac{1}{M_T} \sum_{i=1}^n M_i \omega_i$$

$$M_i = \frac{2H_i}{2\pi f} \quad M_T = \sum_{i=1}^n M_i$$

B. SIME Technique[4]

The SIME technique is based on finding single-machine equivalent of the multi-machine system [3][4]. SIME belongs to the general class of OMIB equivalent. It relies on the observation that loss of synchronism in a power system originates from the irrevocable separation of machines into two groups: a critical machines group (C) and a non-critical machines group (N). These two groups are then replaced by two machines and then by OMIB. Several methods are available for identification of the two groups of machines. In this paper a maximum inertia-weighted angle gap criterion (*Mδ-criterion*) is used. The expressions for transformation of multi-machine system into a SIME based candidate OMIB defined by its angle δ , speed ω , mechanical power P_m , electrical power P_e and inertia coefficient M are as follows.

The COA for critical machines, C:

$$\delta_C(t) = \frac{1}{M_C} \sum_{k \in C} M_k \delta_k(t) \dots\dots (4)$$

The COA for non-critical machines, N:

$$\delta_N(t) = \frac{1}{M_N} \sum_{j \in N} M_j \delta_j(t) \dots\dots (5)$$

The rotor angle of OMIB is given by

$$\delta_{OMIB} = \delta_C(t) - \delta_N(t) \dots\dots (6)$$

Similarly equivalent OMIB rotor speed is expressed as:

$$\omega_C(t) = \frac{1}{M_C} \sum_{k \in C} M_k \omega_k(t) \dots\dots (7)$$

$$\omega_N(t) = \frac{1}{M_N} \sum_{j \in N} M_j \omega_j(t) \dots\dots (8)$$

$$\omega_{OMIB} = \omega_C(t) - \omega_N(t) \dots\dots (9)$$

The equivalent OMIB mechanical and electrical power are given by

$$P_m(t) = M \left(\frac{1}{M_C} \sum_{k \in C} P_{mk}(t) - \frac{1}{M_N} \sum_{j \in N} P_{mj}(t) \right) \dots\dots (10)$$

$$P_e(t) = M \left(\frac{1}{M_C} \sum_{k \in C} P_{ek}(t) - \frac{1}{M_N} \sum_{j \in N} P_{ej}(t) \right) \dots\dots (11)$$

$$M_C = \sum_{k \in C} M_k; \quad M_N = \sum_{j \in N} M_j; \quad M = \frac{M_C M_N}{M_C + M_N}$$

The net accelerating power of the OMIB is

$$P_a(t) = P_m(t) - P_e(t) \dots\dots\dots (12)$$

C. MOD determination ($M\delta$ -criterion) [12]

The $M\delta$ -criterion is used for identification of critical and non-critical group of machines. The maximum inertia-weighted angle gap criterion is applied at the fault clearing time of t_{cl} . The machines are sorted in descending order according to the $M\delta$ product. i.e., $M\delta = M_i \cdot \delta_i(t_{cl})$. The maximum gap between sorted $M\delta_i$ values is identified and machines before the gap forms the critical group (C), whereas the machines after the gap forms non-critical group (N).

IV. CCT EVALUATION BY PROPOSED METHOD

A. Bisection Method

The bisection method in mathematics is used to reduce the computation time required for finding the root of an equation. Initially take an interval $[a, b]$ such that $f(a)$ and $f(b)$ have opposite signs, find the midpoint of $[a, b]$, and then decide whether the root lies on $[a, (a + b)/2]$ or $[(a + b)/2, b]$. Repeat until the interval is sufficiently small. The same concept can be used for reducing the computation time required for CCT evaluation. As shown in figure 4, an initial fault clearing time is assumed ($FCT=10$ sec) and then the limits are chosen such that the system is stable at lower value t_1 and unstable at higher value t_2 . Once again the stability of the system is evaluated at an average value or mid-point t_{mid} . If the system is stable, the average value (t_{mid}) is replaced by lower limit (t_1). Otherwise, the average value (t_{mid}) is replaced by higher value (t_2) in the next calculation. The above procedure is repeated until the bisected time interval becomes less than the specified tolerance ϵ . Then the lower value t_1 is the required CCT.

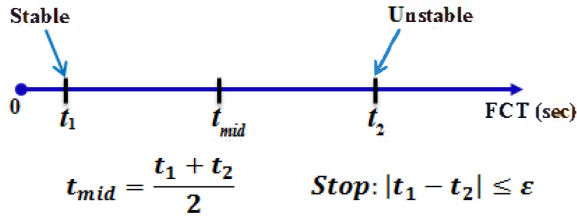


Fig. 4. Illustration of Bisection method used for CCT evaluation

B. CCT Evaluation by Proposed Method

The steps involved in the process of CCT evaluation are given below:

- Step 1: Solve the initial load flow to determine the initial bus voltage magnitudes and phase angles.
- Step 2: Run the conventional TDS with initial fault clearing time.
- Step 3: After each time step of integration obtain the equivalent OMIB rotor speed of the multi-machine system using equations (7) to (9).

Step 4: Starting with the initial fault clearing time find t_1 and t_2 such that at t_1 the system is stable and at t_2 the system is unstable by repeating steps 2 & 3. Use OMIB omega curve for assessing the stability as described in section II.

Step 5: Calculate the mid point t_{mid} and check for stability of the system at t_{mid} .

Step 6: If the system is found stable at t_{mid} , then replace t_{mid} by t_1 or if the system is found unstable at t_{mid} replace t_{mid} by t_2 .

Step 7: Again with the replaced value of fault clearing time continue to bisect the interval until the difference between t_1 and t_2 becomes less than or equal to ϵ .

Step 8: Stop the simulation when the above said criterion is reached. The fault clearing time t_1 will be the required CCT.

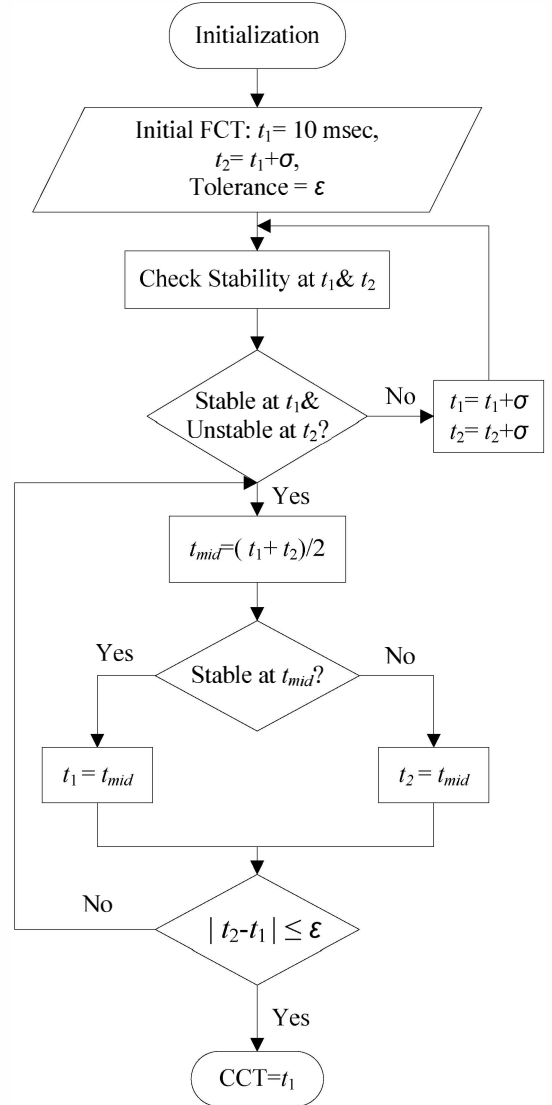


Fig. 5. Estimation of CCT using Bisection method

Figure 5 gives the flowchart of the methodology for calculation of CCT using the integrated approach of bisection and SIME method. In this paper a starting fault clearing time of 10 msec (which is a stable case), $\sigma = 100$ msec and $\varepsilon = 1$ msec is used for CCT estimation.

V. SIMULATION AND RESULTS

The proposed method for estimation of CCT is tested on different test systems of varying size. The CCT obtained by time domain simulation method and by the proposed method have been compared. The results obtained by the proposed method are fairly matching with the CCT values obtained by time domain simulation method. The proposed method is used for CCT evaluation of WSCC 9 Bus, IEEE 14 Bus, IEEE 30 Bus and IEEE 39 Bus New England test systems. The estimated values of critical clearing time and the execution time taken for estimation of CCT for few contingencies of the above mentioned test systems are as given in tables I to IV respectively.

TABLE I. ESTIMATED CCT VALUES AND THE EXECUTION TIME FOR WSCC 9 BUS TEST SYSTEM

Faulted Bus	Removed Line	TDS Method		Proposed Method	
		CCT (sec)	Time (sec)	CCT (sec)	Time (sec)
7	5-7	0.162	5.520	0.162	0.196
9	9-6	0.212	6.814	0.212	0.247
7	7-8	0.181	5.515	0.181	0.206
6	9-6	0.389	13.093	0.389	0.234
4	4-6	0.308	11.389	0.308	0.236

TABLE II. ESTIMATED CCT VALUES AND THE EXECUTION TIME FOR IEEE 14 BUS TEST SYSTEM

Faulted Bus	Removed Line	TDS Method		Proposed Method	
		CCT (sec)	Time (sec)	CCT (sec)	Time (sec)
6	5-6	0.369	13.744	0.368	0.302
7	4-7	0.350	12.868	0.349	0.318
5	4-5	0.129	4.954	0.127	0.257
7	7-9	0.353	12.405	0.349	0.309
9	9-14	0.425	15.797	0.424	0.341

TABLE III. ESTIMATED CCT VALUES AND THE EXECUTION TIME FOR IEEE 30 BUS TEST SYSTEM

Faulted Bus	Removed Line	TDS Method		Proposed Method	
		CCT (sec)	Time (sec)	CCT (sec)	Time (sec)
4	4-6	0.103	4.778	0.102	0.333
6	6-9	0.113	5.244	0.112	0.342
6	6-28	0.114	5.124	0.112	0.393
12	12-14	0.350	17.310	0.346	0.336
28	8-28	0.194	8.902	0.193	0.409

The first column indicates the faulted bus number and the second column gives the line number of the connected line

tripped. The third and fourth columns respectively give the estimated CCT value and the corresponding execution time by conventional time domain simulation method for CCT evaluation. The fifth and sixth columns give the estimated value of CCT and corresponding execution time for CCT evaluation by the proposed method. Here the execution time is the run time for computation of CCT for single contingency.

TABLE IV. ESTIMATED CCT VALUES AND THE EXECUTION TIME FOR IEEE 39 BUS TEST SYSTEM

Faulted Bus	Removed Line	TDS Method		Proposed Method	
		CCT (sec)	Time (sec)	CCT (sec)	Time (sec)
1	1-2	0.240	13.269	0.240	0.557
5	4-5	0.207	11.155	0.208	0.280
8	8-9	0.199	10.004	0.199	0.256
10	10-13	0.202	10.418	0.202	0.288
13	13-14	0.211	10.780	0.212	0.285
17	16-17	0.137	7.477	0.140	0.234
27	17-27	0.180	10.265	0.183	0.249
39	1-39	0.293	15.275	0.293	0.274

For CCT evaluation using conventional TDS method, simulation is started by clearing the fault immediately and the stability is checked by inspection of angular difference. Then the fault clearing time is increased until the instability is detected. For example for a three phase fault at bus number 4, of the WSCC 9 Bus system, the line connected between buses 4 and 6 is tripped for clearing the fault. The corresponding CCT value is found to be 0.308 seconds. The execution time taken for CCT evaluation by proposed method is 0.236 seconds, whereas the execution time taken by TDS method is 11.389 seconds. From the comparison of execution time taken by both the methods it is observed that time taken for CCT evaluation using proposed method is very less than the conventional TDS method. This indicates the computational speed of the proposed method. The simulation is carried out in MATLAB. The computer configuration used for simulation is Intel(R), Core i5, 2.5-GHz CPU with 4 GB of RAM.

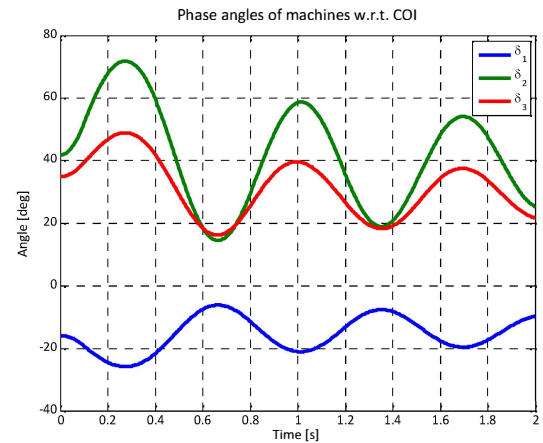


Fig. 6. Rotor angles of the machines w.r.t COI for a stable case

Figures 6 to 8 shows respectively the plots of rotor angles of the machines w.r.t. COI reference frame, rotor angle delta of the equivalent OMIB and omega of the OMIB for a stable case. The fault is applied to bus 7 of WSCC 9 bus system and is cleared by opening the connected line 5-7 at 100 msec, which is a stable case corresponding to CCT of 162 msec.

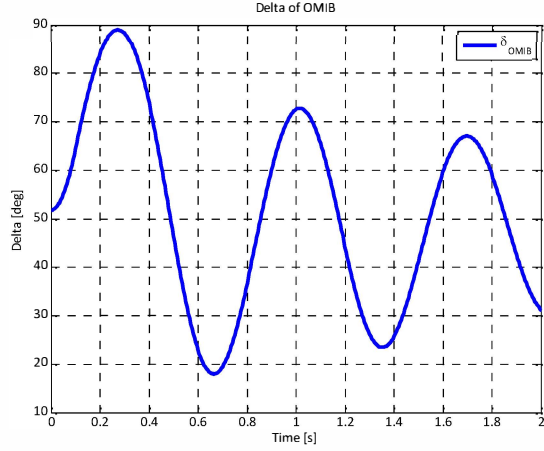


Fig. 7. Delta of OMIB for a stable case

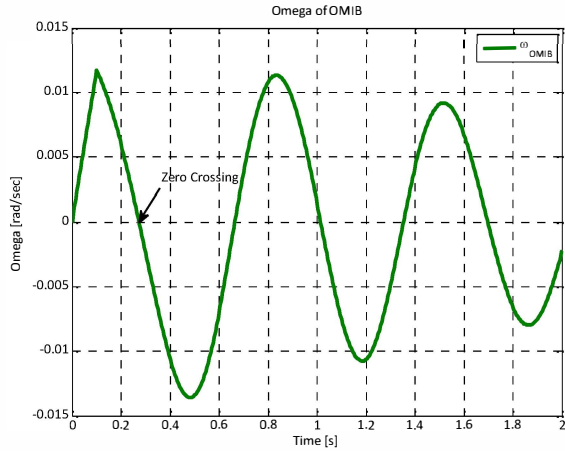


Fig. 8. Omega of OMIB for a stable case

The plots of rotor angles of machines w.r.t COI reference frame, delta of the equivalent OMIB and omega of the OMIB for an unstable case corresponding to the fault clearing time of 180 msec are as shown in figures 9 to 11. It can be observed from the figures 8 and 11 that for a stable case, the omega of the equivalent OMIB will cross the zero-axis indicating the stable condition of the system after the fault clearance. For an unstable case the omega of the OMIB will not cross the zero - axis, instead it starts increasing after the fault clearance indicating unstable condition of the system. In case of OMIB, after the fault clearance stable or unstable condition of the system can be judged early and easily by using the above mentioned criterion compared to conventional TDS method. As soon as the stability or instability is detected the simulation can

be stopped saving the computation time. This is the novelty of the proposed method.

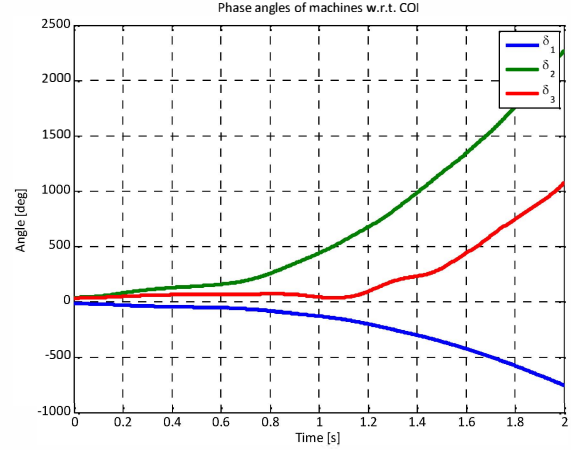


Fig. 9. Rotor angles of the machines w.r.t COI for an unstable case

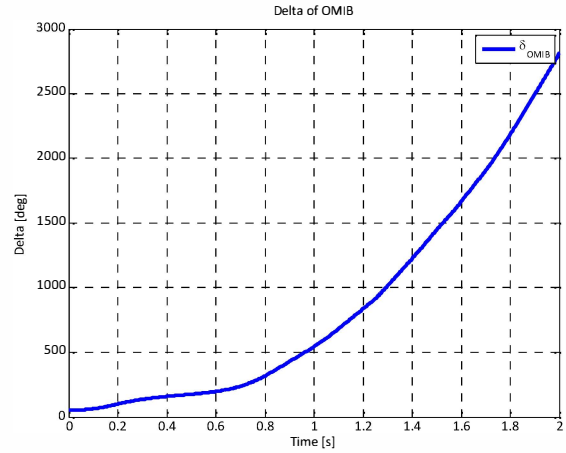


Fig. 10. Delta of OMIB for an unstable case

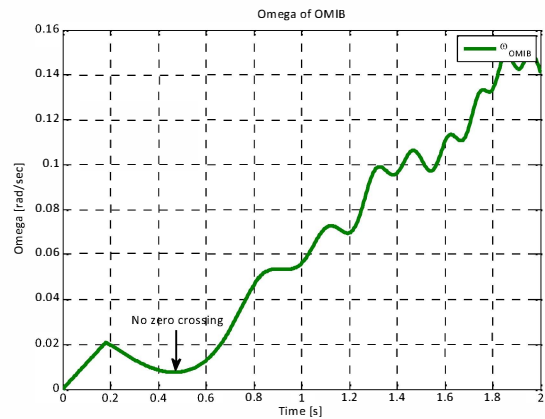


Fig. 11. Omega of OMIB for an unstable case

VI. CONCLUSIONS

A methodology for critical clearing time estimation of multi-machine systems has been presented in this paper. The proposed method has the advantages of both SIME and TDS methods. The novelty of the proposed method is its fastness to detect stability and accurate evaluation of CCT. Comparison of the results indicates the fast computation of the CCT by proposed method as compared to conventional time domain simulation method.

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